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1 D:\Software\Python\Python311\python.exe E:\PythonProjectsByYear\Year2025\MC-
  PIKAN_Official\experiments\Volterra1D_MCPI.py
2 The network model is PINNs
3 Layers: ModuleList(
4   (0): Linear(in_features=1, out_features=20, bias=True)
5   (1): Linear(in_features=20, out_features=20, bias=True)
6   (2): Linear(in_features=20, out_features=1, bias=True)
7 )
8 Activation function: Tanh()
9 Total trainable parameters: 481
10 Model on device: cuda:0
11 Start training...
12 Max Epochs: 2000, Learning Rate: 0.001, Epsilon: 1e-20, Optimizer: Adam
13 Training: 100%|██████████| 2000/2000 [00:06<00:00, 330.31epoch/s, Loss=1.
  39e-04, LR=1.0e-03, Best=1.39e-04]
14 Training finished. Best loss: 1.06e-04. Model saved to results/best_model.pth
15 Loss history saved to results/loss_history.npy
16 Training time: 6.846659421920776s
17 Relative loss: 0.06078052148222923
18 L2 loss: 0.0386747308075428
19 Parameter containing:
20 tensor([[ -1.2408],
21         [ -0.1472],
22         [  0.8108],
23         [ -1.1641],
24         [ -1.7066],
25         [  0.3443],
26         [ -0.5842],
27         [  1.4643],
28         [  0.9901],
29         [ -0.5635],
30         [ -1.2671],
31         [ -0.5945],
32         [  1.5172],
33         [ -0.0657],
34         [  0.0478],
35         [ -0.5966],
36         [ -0.4332],
37         [ -0.3124],
38         [  1.2543],
39         [ -0.7627]], device='cuda:0', requires_grad=True)
40 Parameter containing:
41 tensor([ 1.0528, -0.1534, -0.2409,  0.8417,  1.4692, -0.4182, -0.1219,  0.1850,
42         -0.1947, -0.5362, -0.6380, -0.2365, -0.4005,  0.6682,  0.9369, -0.3750,
43         -0.1916, -0.4879,  0.1756, -0.6849], device='cuda:0',
44         requires_grad=True)
45 Parameter containing:
46 tensor([[ 3.6396e-01, -2.0277e-01,  4.5467e-02,  4.0297e-01,  3.4299e-01,
47         6.5338e-02,  1.9345e-01, -1.6799e-02, -1.4376e-01,  1.3333e-01,
48         3.2352e-02,  2.1508e-02, -7.2182e-02,  2.1873e-01,  1.1353e-01,
49         -1.0716e-02, -1.2659e-01,  4.6765e-02, -1.3004e-01,  1.9917e-02],
50         [-8.6384e-01, -1.3272e-01,  6.5006e-01, -1.2788e+00, -1.0683e+00,
51         2.3178e-01, -1.8113e-01, -2.0340e-02,  4.2224e-01,  2.3530e-01,

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52	1.5151e-01, -2.4230e-01, -1.4908e-01, -2.0405e-01, -2.0988e-01,
53	2.4488e-02, -1.7807e-01, -1.2811e-01, -1.9637e-01, 1.1803e-01],
54	[2.6841e-01, -1.0311e-01, -2.1846e-01, 4.1994e-01, -2.6211e-02,
55	9.5585e-02, 2.9782e-02, 2.0345e-01, -6.9727e-02, 8.8221e-02,
56	9.6285e-02, 1.1524e-01, -1.1365e-01, 5.8947e-02, 1.1693e-01,
57	-1.9598e-01, 9.8530e-02, 5.5638e-02, -1.5912e-02, -3.0384e-02],
58	[-2.6925e-03, -2.7210e-01, 1.1454e-03, -3.7281e-01, -2.0148e-01,
59	-1.4788e-01, -1.3145e-01, 1.6374e-01, 1.6208e-01, 1.1751e-01,
60	1.4061e-01, -1.9968e-01, -2.2751e-01, 4.4999e-02, -1.2839e-01,
61	1.6070e-01, 1.1463e-01, 1.2909e-03, 6.4671e-02, -1.3676e-01],
62	[1.9276e-01, 3.4332e-02, -4.0810e-01, 6.9054e-01, 4.6516e-01,
63	-2.8280e-01, 2.2764e-01, -1.0168e-01, -4.5866e-01, 1.4071e-01,
64	1.9670e-01, 2.5042e-01, -1.8481e-01, 8.0014e-03, -5.7350e-02,
65	2.0171e-01, -7.2187e-02, -1.2866e-01, 8.6273e-02, 1.3308e-02],
66	[8.4619e-02, 3.5960e-02, -9.0886e-02, -2.2586e-02, 8.8020e-02,
67	1.9092e-03, -1.9964e-02, -8.6419e-02, -2.4488e-01, 7.1653e-02,
68	-5.2126e-02, -1.6645e-01, 1.5278e-01, 9.4439e-02, -6.5414e-03,
69	-1.0201e-01, -1.9572e-02, 1.5263e-01, -8.2593e-02, 1.2591e-01],
70	[-6.5843e-02, 1.2083e-01, -1.5884e-01, -1.0956e-01, -3.3181e-01,
71	1.8754e-01, 6.3475e-02, 1.6459e-01, 4.4150e-02, 1.9251e-01,
72	2.0069e-01, -6.5807e-02, 1.4907e-01, 9.9028e-02, -1.3088e-01,
73	7.3276e-02, 1.9907e-02, 8.0703e-02, 5.3646e-02, -1.1164e-02],
74	[-9.8240e-02, 1.6804e-01, -7.2743e-02, 1.4634e-01, 2.9886e-01,
75	2.5227e-01, 7.0308e-02, -3.1356e-03, -1.0949e-02, 2.1451e-01,
76	1.3076e-01, 3.9462e-02, 2.6345e-01, -3.8986e-02, 1.4451e-01,
77	-1.1202e-01, 7.0906e-03, 2.7204e-01, 2.1662e-01, 8.7255e-02],
78	[-8.0450e-02, -1.4832e-01, -5.6022e-02, 8.7072e-02, -2.3153e-01,
79	5.2942e-02, 1.4447e-01, 1.4444e-01, -8.0448e-02, -2.2054e-01,
80	-4.3228e-02, -1.3979e-02, -1.6014e-01, 2.2355e-01, -5.0791e-02,
81	-2.5256e-02, -1.7374e-01, -2.1136e-02, -1.2464e-01, -1.4934e-01],
82	[3.2098e-01, 1.0730e-01, -5.7939e-01, 3.6836e-01, 3.4721e-01,
83	-1.0408e-01, 1.1648e-01, -8.1070e-03, -2.1689e-01, 1.6381e-01,
84	9.7115e-02, 1.6887e-01, -3.6647e-01, -1.5484e-01, -8.7261e-02,
85	1.8728e-01, 9.1375e-02, 2.2413e-01, 6.3546e-02, 6.6384e-02],
86	[3.7190e-01, 1.3032e-01, -1.9663e-01, 9.1241e-01, 5.1124e-01,
87	-3.9017e-01, 1.0661e-01, 2.2454e-01, -8.6098e-02, 5.0367e-02,
88	-1.5615e-01, 7.2717e-02, -1.5755e-01, 1.1046e-01, 1.0728e-02,
89	8.6924e-02, 2.0281e-01, -5.9123e-02, 8.4853e-02, -5.1744e-02],
90	[1.3830e-02, -2.4999e-02, -2.7575e-03, -4.2829e-02, -7.9173e-02,
91	-6.8254e-02, -1.0754e-01, 1.3756e-01, -2.1847e-02, 2.2483e-01,
92	1.3278e-01, 7.7762e-02, 1.0465e-02, 7.1784e-02, -3.6887e-02,
93	1.0772e-01, -6.8702e-02, 7.9740e-04, -1.3839e-01, -1.8285e-02],
94	[-1.0005e-01, -9.6541e-02, -1.4455e-01, 7.8388e-02, 8.6435e-02,
95	-1.6506e-02, 1.1490e-01, 3.3669e-02, -1.0887e-01, 9.1824e-02,
96	-6.8806e-02, 5.4566e-02, 1.3642e-01, 1.4536e-02, 1.5929e-01,
97	4.5177e-02, 2.5520e-01, -7.5554e-02, 1.7489e-01, -4.2863e-02],
98	[6.9197e-02, -2.6381e-01, 1.8645e-02, 4.5561e-03, 8.2990e-02,
99	-2.2314e-01, -2.3591e-01, -1.6480e-01, 6.4731e-02, -6.7442e-03,
100	-2.0586e-01, -5.1136e-02, 5.6294e-02, -1.4365e-01, -1.0705e-01,
101	1.5594e-01, -1.0552e-01, -1.7126e-01, -3.1230e-02, -2.1856e-01],
102	[-5.2336e-01, -4.5400e-01, 6.5233e-01, -6.0116e-01, -5.1798e-01,
103	4.6005e-01, -6.4790e-01, 5.7883e-01, 8.3962e-01, -5.0947e-01,
104	-3.2445e-01, -9.3691e-02, 6.3390e-01, 2.0858e-01, 3.0849e-02,

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105     -4.2710e-01, -3.2835e-01, -1.3122e-01,  5.9800e-01, -3.9011e-01],
106     [-4.2100e-02, -8.3958e-02,  1.1070e-01, -2.9675e-01, -1.0964e-02,
107     2.9763e-01, -3.5823e-02,  2.4540e-01,  3.2100e-01,  2.1607e-02,
108     -1.0488e-01, -1.4862e-01,  1.8077e-01, -1.2610e-01, -1.9400e-01,
109     -6.7595e-02,  1.8779e-01, -1.6052e-02, -6.9978e-03, -2.2270e-01],
110     [ 1.0334e-01, -1.4213e-01, -1.8228e-01, -1.9949e-01, -7.9699e-02,
111     -6.4582e-02,  1.5700e-01, -1.4657e-01,  1.3876e-02, -2.5105e-01,
112     -8.1071e-02, -2.6361e-01,  5.0660e-02,  2.0335e-01,  2.7661e-01,
113     1.0601e-01,  9.1090e-02, -1.4898e-02, -3.8877e-02,  1.8761e-02],
114     [ 6.7451e-02, -8.1343e-02, -4.3213e-01,  5.6333e-01,  3.4679e-01,
115     -2.4204e-01,  1.7496e-01, -1.6154e-01, -1.1038e-01,  1.7915e-01,
116     2.5533e-02,  1.4203e-01, -2.3814e-01,  3.8921e-02,  4.0267e-02,
117     -6.7279e-02,  1.7806e-01, -2.0172e-01, -9.5823e-02,  2.0800e-01],
118     [-1.6218e-01,  3.2877e-02, -8.1113e-02, -1.3990e-01, -1.1931e-01,
119     1.4075e-02, -1.4646e-01,  7.2688e-02,  1.4621e-01, -1.9013e-01,
120     1.1689e-01,  4.8026e-02, -2.6270e-01,  3.6648e-02, -7.8488e-02,
121     9.6464e-02, -2.3320e-01, -2.0650e-01, -1.3102e-01, -1.7122e-01],
122     [ 2.8038e-01, -9.9243e-03,  2.2571e-02,  7.3623e-01,  1.1587e-01,
123     -2.5378e-01,  1.4016e-01, -8.2355e-02, -1.8994e-01, -1.6100e-02,
124     -1.8130e-01,  6.8234e-02, -2.1916e-01,  2.1753e-01,  1.7478e-01,
125     2.2516e-01, -5.6439e-02, -1.6606e-01, -4.9844e-03,  5.7988e-02]],
126     device='cuda:0', requires_grad=True)
127 Parameter containing:
128 tensor([ 0.2450, -0.2286, -0.0326, -0.0786, -0.1735,  0.1263,  0.1681, -0.0200,
129         0.1870,  0.1777,  0.1504, -0.2228, -0.2323,  0.1882, -0.1813, -0.0545,
130         0.0744, -0.1577,  0.0769,  0.2215], device='cuda:0',
131         requires_grad=True)
132 Parameter containing:
133 tensor([[ 0.2488, -0.1352,  0.1488, -0.1435, -0.1503, -0.0061, -0.0854,  0.2849,
134         -0.1580, -0.2075,  0.4993,  0.1086,  0.0507, -0.1899,  0.3923,  0.2293,
135         -0.2213, -0.1089, -0.3727,  0.1357]], device='cuda:0',
136         requires_grad=True)
137 Parameter containing:
138 tensor([-0.0865], device='cuda:0', requires_grad=True)
139 =====
140 Layer (type:depth-idx)          Output Shape          Param #
141 =====
142 PINNModel                      [1, 1]                --
143 |—ModuleList: 1-5              --                      (recursive)
144 |   |—0.weight                  |—20
145 |   |—0.bias                    |—20
146 |   |—1.weight                  |—400
147 |   |—1.bias                    |—20
148 |   |—2.weight                  |—20
149 |   |—2.bias                    |—1
150 |   |—Linear: 2-1              [1, 20]              40
151 |   |   |—weight                |—20
152 |   |   |—bias                  |—20
153 |—Tanh: 1-2                    [1, 20]              --
154 |—ModuleList: 1-5              --                      (recursive)
155 |   |—0.weight                  |—20

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156 |      | 0.bias | 20
157 |      | 1.weight | 400
158 |      | 1.bias | 20
159 |      | 2.weight | 20
160 |      | 2.bias | 1
161 |      | Linear: 2-2 | [1, 20] | 420
162 |      | | weight | 400
163 |      | | bias | 20
164 | Tanh: 1-4 | [1, 20] | --
165 | ModuleList: 1-5 | -- | (recursive)
166 |      | 0.weight | 20
167 |      | 0.bias | 20
168 |      | 1.weight | 400
169 |      | 1.bias | 20
170 |      | 2.weight | 20
171 |      | 2.bias | 1
172 |      | Linear: 2-3 | [1, 1] | 21
173 |      | | weight | 20
174 |      | | bias | 1
175 | =====
176 |      |
177 | Total params: 481
178 | Trainable params: 481
179 | Non-trainable params: 0
180 | Total mult-adds (Units.MEGABYTES): 0.00
181 | =====
182 |      |
183 | Input size (MB): 0.00
184 | Forward/backward pass size (MB): 0.00
185 | Params size (MB): 0.00
186 | Estimated Total Size (MB): 0.00
187 | =====
188 |      |
189 | 进程已结束, 退出代码为 0

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